

A tensor-hypercontraction, Laplace-transform formulation of second-order Møller–Plesset theory for two-component relativistic reference determinants with a bounded memory footprint

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(Dated: 31 August 2026)

We describe a formulation and implementation of second-order Møller–Plesset perturbation theory (MP2) for two-component, spin–orbit-coupled reference determinants (exact two-component molecular mean-field, X2CAMF) in which no four-index quantity is ever constructed. The antisymmetrised correlation energy is separated into its direct (Coulomb) and exchange Goldstone diagrams. The direct diagram is contracted through a grid tensor-hypercontraction (THC) representation of the electron-repulsion integrals combined with a Laplace transform of the doubles denominator; the transform factorises the four orbital labels, and the diagram collapses onto a sequence of $K \times K$ grid-space matrix multiplications, where K is the number of retained interpolation points. The exchange diagram, whose index pattern the Laplace transform does not decouple and which is therefore $\mathcal{O}(K^4)$ under THC, is instead evaluated by the resolution of the identity with the occupied labels kept explicit; because those labels are explicit no quadrature is required and the exact orbital-energy denominator is used, so the exchange contribution carries no Laplace error. The interpolation points and the THC core are built by a grid-blocked interpolative separable density fitting procedure that never forms an $N_{\text{grid}} \times N_{\text{grid}}$ or $N_{\text{grid}} \times n_{\text{mo}}^2$ array. The direct kernel has an in-core path and an out-of-core path in which only two $K \times K$ matrices are resident and the THC core is streamed in row panels, so the peak memory is $\mathcal{O}(K^2)$ or $2K^2 + \mathcal{O}(K n_{\text{mo}})$ rather than $\mathcal{O}(o^2 v^2)$ or $\mathcal{O}(N_{\text{aux}} n_{\text{mo}}^2)$. Spin-component scaling of the two diagrams reproduces the spin-component-scaled and scaled-opposite-spin variants; with the exchange diagram switched off the method is a scaled-opposite-spin relativistic MP2 whose cost is $\mathcal{O}(K^3)$ per Laplace node. The implementation, its compiled kernels, and its validation against a dense density-fitted MP2 in the same spinor molecular orbital basis are described.

I. INTRODUCTION

Second-order Møller–Plesset perturbation theory is the least expensive size-consistent wave-function correlation method and the entry point of the correlation hierarchy. Its canonical cost is $\mathcal{O}(o^2 v^2 N_{\text{aux}})$ in a density-fitted (resolution-of-the-identity, RI) formulation,^{1–4} with o and v the numbers of occupied and virtual orbitals; the memory cost, if the doubles amplitudes or the $(ov|ov)$ integral block are held, is $\mathcal{O}(o^2 v^2)$. For a two-component relativistic treatment on a spinor basis both counts refer to complex arithmetic and to a spinor dimension roughly twice the number of scalar atomic-orbital basis functions, so the $\mathcal{O}(o^2 v^2)$ memory is the binding constraint well before the arithmetic is: at a few thousand scalar basis functions the four-index amplitude is measured in terabytes and the three-index RI tensor in hundreds of gigabytes.

Two ideas reduce this. The Laplace transform of the energy denominator,^{5–7} $1/x = \int_0^\infty e^{-xt} dt \approx \sum_g w_g e^{-x t_g}$, separates the four orbital energies that enter an MP2 denominator into a product of per-orbital exponential factors; those factors can be absorbed into the orbitals, after which the direct (Coulomb) part of the correlation energy becomes a trace over a product of two-index quantities.^{8,9} Tensor hypercontraction^{11–13} represents the electron-repulsion integral as a contraction of grid collocation matrices X_{Pp} (orbital p evaluated at grid point P) with a small core Z_{PQ} ,

$$(pq|rs) = \sum_{PQ} \rho_{pq}(P) Z_{PQ} \rho_{rs}(Q), \quad \rho_{pq}(P) = \sum_k X_{kPp}^* X_{kPq}, \quad (1)$$

so that any integral network can be rewritten in the K -dimensional grid space. The interpolation points and the core are obtained by least squares against the RI integrals; a robust and inexpensive route to the points is interpolative separable density fitting (ISDF),^{14,15} which selects them by a pivoted Cholesky (equivalently a column-pivoted QR) factorisation of the collocation Gram matrix.

Combined, THC and the Laplace transform give an MP2 direct energy at $\mathcal{O}(K^3)$ per Laplace node with only $\mathcal{O}(K^2)$ storage.^{8,16,17} The exchange energy is the difficulty: its integrand, $(ia|jb)(ib|ja)$, ties the occupied label i to the virtual labels of *both* factors, so no rearrangement of the Laplace product separates the four grid sums, and a THC evaluation is irreducibly $\mathcal{O}(K^4)$ (or $\mathcal{O}(o K^2 v)$ with the occupied labels retained), with a prefactor $K/n_{\text{mo}} \sim 10$ that makes it impractical past a thousand or so functions. We therefore

evaluate only the direct diagram by THC and Laplace, and the exchange diagram by RI with the occupied labels kept explicit — a hybrid that keeps the memory bounded and, because the exchange labels are explicit, keeps the exchange contribution free of quadrature error.

The target is the exact two-component molecular-mean-field (X2CAMF) Hamiltonian,^{21,22} for which BAGH already provides Cholesky-decomposed spinor integrals and a THC representation used by its coupled-cluster codes. The present method reuses that machinery. What follows is the working equations (Sec. II) and the implementation, including the grid-blocked ISDF build, the in-core and out-of-core direct kernels, the screened exchange kernel, and frozen-core handling (Sec. III); Sec. IV collects the costs and Sec. V the numerical checks.

II. THEORY

A. The two diagrams

In a spinor basis the MP2 correlation energy for a single-determinant reference is

$$E_{\text{MP2}} = -\frac{1}{4} \sum_{ijab} \frac{|\langle ij||ab \rangle_A|^2}{D_{ijab}}, \quad \langle ij||ab \rangle_A = (ia|jb) - (ib|ja), \quad (2)$$

with $D_{ijab} = \varepsilon_a + \varepsilon_b - \varepsilon_i - \varepsilon_j$, indices i, j occupied and a, b virtual, $(ia|jb)$ the chemists' electron-repulsion integral over spinors, and ε the (semi-) canonical spinor energies. Expanding the antisymmetriser,

$$|(ia|jb) - (ib|ja)|^2 = |(ia|jb)|^2 + |(ib|ja)|^2 - 2 \operatorname{Re}[(ia|jb)(ib|ja)^*], \quad (3)$$

and using that the first two terms are equal under $i \leftrightarrow j$, $a \leftrightarrow b$ relabelling of the sum,

$$E_{\text{MP2}} = E_{\text{dir}} + E_{\text{exc}}, \quad (4)$$

$$E_{\text{dir}} = -\frac{1}{2} \sum_{ijab} \frac{|(ia|jb)|^2}{D_{ijab}}, \quad (5)$$

$$E_{\text{exc}} = +\frac{1}{2} \sum_{ijab} \frac{\operatorname{Re}[(ia|jb)(ib|ja)^*]}{D_{ijab}}. \quad (6)$$

Both sums have $D_{ijab} > 0$ term by term. The exchange sum equals its own complex conjugate under $i \leftrightarrow j$ together with the permutational symmetry $(pq|rs) = (rs|pq)$, so E_{exc} is real and the Re in Eq. (6) may be taken inside.

B. Direct diagram: THC and Laplace

Write the Coulomb integral in the grid THC form Eq. (1), with the collocation split into occupied and virtual blocks, X_{kPi}^o and X_{kPa}^v , and k running over the two spinor components. Replace the denominator by the Laplace quadrature, $1/D_{ijab} = \sum_g w_g e^{-D_{ijab}t_g}$, valid because D_{ijab} lies in a bounded positive interval; the geometric (double-exponential in $1/h$) quadrature of Ref. 10 on $[2(\varepsilon_v^{\min} - \varepsilon_o^{\max}), 2(\varepsilon_v^{\max} - \varepsilon_o^{\min})]$ is used, the factor two being the doubles analogue of the factor three appropriate to triples. The Boltzmann factor factorises as $e^{-D_{ijab}t} = e^{(\varepsilon_i - \varepsilon_a)t} e^{(\varepsilon_j - \varepsilon_b)t}$; splitting each half evenly and folding it into the collocation,

$$Y_{kPi}^o(t) = X_{kPi}^o e^{+\varepsilon_i t/2}, \quad Y_{kPa}^v(t) = X_{kPa}^v e^{-\varepsilon_a t/2}, \quad (7)$$

so that the “dressed” pair density $\tilde{\rho}_{ia}(P) = \sum_k Y_{kPi}^{o*} Y_{kPa}^v$ carries $e^{(\varepsilon_i - \varepsilon_a)t/2}$. Substituting Eqs. (1),(7) into Eq. (5) and summing the orbital labels,

$$E_{\text{dir}} = -\frac{1}{2} \sum_g w_g \sum_{PR} M_g[P, R] (Z M_g Z^\dagger)[P, R], \quad (8)$$

$$\begin{aligned} M_g[P, R] &= \sum_{ia} \tilde{\rho}_{ia}(P) \tilde{\rho}_{ia}(R)^* \\ &= \sum_{kk'} \left(\sum_i Y_{kPi}^{o*} Y_{k' Ri}^o \right) \left(\sum_a Y_{kPa}^v Y_{k' Ra}^{v*} \right). \end{aligned} \quad (9)$$

M_g is a Hermitian $K \times K$ matrix, assembled as a Hadamard product of two $K \times K$ Gram matrices for each of the four spinor-component pairs (k, k') : eight complex matrix multiplications of shapes $(K \times o)(o \times K)$ and $(K \times v)(v \times K)$. The bracketed quantity in Eq. (8) is $N_g = Z M_g Z^\dagger$, two further $K \times K$ multiplications, and the node’s contribution is the Frobenius inner product $\sum_{PR} M_g[P, R] N_g[P, R]$, which is real and non-negative. The cost per node is $\mathcal{O}(K^3)$; the largest array is $K \times K$.

C. Exchange diagram: RI with exact denominators

Because the Laplace product for the exchange integrand does not separate the grid sums, we keep the occupied labels explicit and use the RI factorisation $(ia|jb) = \sum_L L_{ia}^L L_{jb}^L$ that the reference already supplies (the same three-index quantity from which the THC core is fitted). Define, for an ordered occupied pair (i, j) ,

$$T_{ab}^{ij} = \sum_L L_{ia}^L L_{jb}^L, \quad (10)$$

a $v \times v$ matrix formed by one contraction over the auxiliary index. Then Eq. (6) is

$$E_{\text{exc}} = \frac{1}{2} \sum_{i \leq j} m_{ij} \sum_{ab} \frac{\text{Re}[T_{ab}^{ij} (T_{ba}^{ij})^*]}{\varepsilon_a + \varepsilon_b - \varepsilon_i - \varepsilon_j}, \quad (11)$$

with $m_{ij} = 1$ for $i = j$ and $m_{ij} = 2$ otherwise (the pair (j, i) contributes the complex conjugate, with equal real part). No quadrature enters Eq. (11): the denominator is the exact orbital-energy gap, so E_{exc} is exact given the RI fit of the integrals. The cost is one $\mathcal{O}(N_{\text{aux}} v^2)$ contraction and one $\mathcal{O}(v^2)$ weighted trace per pair, $\mathcal{O}(o^2(N_{\text{aux}} + v)v^2)$ in total, with $\mathcal{O}(v^2)$ working memory.

For a spatially extended system with localised occupied orbitals^{19,20} the summand of Eq. (11) decays with the separation of i and j . A Cauchy–Schwarz bound,

$$\|T^{ij}\|_F \leq s_{ij} \equiv \sum_L \|L_i^L\|_2 \|L_j^L\|_2, \quad (12)$$

is evaluated as a single $\mathcal{O}(o^2 N_{\text{aux}})$ matrix multiplication and used to drop pairs with s_{ij} below a threshold times $\max_{ij} s_{ij}$, taking the pair count from $\mathcal{O}(o^2)$ towards $\mathcal{O}(o)$.

D. Spin-component scaling

Under spin–orbit coupling S_z is not a good quantum number, so a rigorous opposite-/same-spin partition of the MP2 energy does not exist. Equations (5)–(6) are the Coulomb and exchange antisymmetrised diagrams; scaling them independently,

$$E_{\text{MP2}} = c_{\text{os}} E_{\text{dir}} + c_{\text{ss}} E_{\text{exc}}, \quad (13)$$

recovers the non-relativistic spin-component-scaled¹⁸ and scaled-opposite-spin⁸ prescriptions in the weak-coupling limit and provides their natural relativistic generalisation otherwise. Setting $c_{\text{ss}} = 0$ removes the exchange diagram entirely; the remaining method is a scaled-opposite-spin relativistic MP2 evaluated purely by Eq. (8), at $\mathcal{O}(K^3)$ per node and $\mathcal{O}(K^2)$ memory.

III. IMPLEMENTATION

The driver is a Python module; the two rate-limiting kernels — Eq. (8) and the pair loop of Eq. (11) — are a compiled PYBIND11 extension that calls level-3 BLAS (`zgemm`) and

is parallelised over Laplace nodes and grid panels, or over occupied pairs, with OpenMP. Reference implementations of both equations in dense array operations are retained and are used automatically when the extension is absent; they are the object of the validation in Sec. V.

A. ISDF factor construction

When the reference is run with the THC option the X2CAMF driver supplies X^o , X^v and Z directly. Otherwise they are built here from the density-fitted integrals without forming any array of grid-squared or grid $\times$ orbital-squared size:

1. The two-component collocation $X_{kPp} = w_P^{1/4} \sum_{\mu} \phi_{\mu}^{(k)}(\mathbf{r}_P) C_{\mu p}$ is evaluated in blocks of grid points, w_P the quadrature weight.
2. Interpolation points are chosen by a pivoted Cholesky factorisation of the collocation Gram matrix $S_{PQ} = \sum_{pq} \rho_P(pq)^* \rho_Q(pq) = \sum_{kk'} |\sum_p X_{kPp} X_{k'Qp}^*|^2$, with the Schur-updated diagonal maintained explicitly and each required column of S recomputed on demand in $\mathcal{O}(N_{\text{grid}} n_{\text{mo}})$ work; no $N_{\text{grid}} \times N_{\text{grid}}$ matrix is ever stored. Selection stops when the updated diagonal falls below a relative floor, so K is rank-adaptive.
3. The collocation is gathered on the K retained points to give X^o and X^v .
4. The THC core solves the least-squares problem $Z = S^{-1} E S^{-1}$ with $E_{PQ} = \sum_L (\sum_{pq} \rho_P(pq) L_{pq}^L) (\sum_{p'q'} \rho_Q(p'q') L_{p'q'}^L)$ the pair densities on the retained points formed in blocks of P and the auxiliary index streamed; E is accumulated as one $K \times N_{\text{aux}}$ panel times its transpose. S is regularised with a small multiple of its trace and inverted by a Cholesky solve (a pseudoinverse with an explicit cutoff as fallback). Z is symmetrised.

For large K the core is written to disk and returned as a lazy dataset. The block sizes are chosen from the available memory; the construction is guarded so that a request whose $K n_{\text{mo}}^2$ working set would not fit is refused with a diagnostic rather than thrashing.

B. Direct kernel: in-core and out-of-core

The kernel loops over Laplace nodes. For each node it forms the dressed collocation Eq. (7), builds M_g from Eq. (9) as eight `zgemm` calls and a Hadamard accumulation, and

evaluates Eq. (8). Three residency regimes are selected automatically from the estimated resident set against a memory budget:

- *in-core*: M_g , N_g and Z are all held; the compiled kernel runs the entire node loop. Peak $\approx 4K^2$ complex plus the collocation.
- *out-of-core*: only M_g and $W_g = M_g Z^\dagger$ are held and Z is read in row panels from the on-disk dataset; $N_g = ZW_g$ and the inner product are accumulated panel by panel. Peak $\approx 2K^2$ complex plus the collocation, i.e. $2K^2 + \mathcal{O}(Kn_{\text{mo}})$.
- a disk-backed Z forces the out-of-core path.

The out-of-core path is pure BLAS-bound array algebra and is not moved into the compiled extension; the in-core path is.

C. Exchange kernel

The Schwarz bound Eq. (12) is formed from per-occupied auxiliary-row norms and the surviving ordered pairs ($i \leq j$) are tiled by blocks of the occupied index so that each three-index slice $L_{i \in \mathcal{I}, a}^L$ that must be resident is bounded. For each tile the compiled kernel loops over its pairs; per pair it forms T^{ij} (Eq. (10)) by one `zgemm` contracting the auxiliary index of the two strided slices, then accumulates $m_{ij} \sum_{ab} \text{Re}[T_{ab}^{ij}(T_{ba}^{ij})^*]/(\varepsilon_a + \varepsilon_b - \varepsilon_i - \varepsilon_j)$ into a thread-private scalar. The dressing of Eq. (7) is *not* applied here — there is no Laplace grid — and no dressed copy of the three-index integral is made.

D. Frozen core

The standard frozen-core keywords select a number of lowest occupied spinors to exclude. The occupied collocation X^o , the occupied energy vector, the three-index integral L_{ia} , and the interval on which the Laplace nodes are placed are all sliced to the active occupied space before either diagram is evaluated; the virtual space and K are unchanged. The frozen core acts identically on both diagrams, and the correlation energy reproduces a dense frozen-core RI-MP2 in the same active space to the THC and quadrature accuracy.

TABLE I. Formal cost and peak working memory. o , v occupied and virtual spinor counts; N_{aux} auxiliary functions; K retained interpolation points ($K \sim c n_{\text{mo}}$, $c \sim 8\text{--}15$); N_{grid} parent grid points; $N_{\text{grid}}^{\text{LT}}$ Laplace nodes; p the P -block size in the ISDF build.

step	cost	peak memory
ISDF points & core	$K N_{\text{aux}} n_{\text{mo}}^2$	$K^2 + p n_{\text{mo}}^2$
direct, in-core	$K^3 N_{\text{grid}}^{\text{LT}}$	$4K^2$
direct, out-of-core	$K^3 N_{\text{grid}}^{\text{LT}}$	$2K^2 + K n_{\text{mo}}$
exchange (RI)	$o^2(N_{\text{aux}} + v) v^2$	v^2

IV. COMPUTATIONAL COST AND MEMORY

Table I collects the counts. No step scales as $o^2 v^2$ or $N_{\text{aux}} n_{\text{mo}}^2$ in memory. The direct diagram is the rate-determining arithmetic at $\mathcal{O}(K^3)$ per node; with the geometric Laplace quadrature $N_{\text{grid}}^{\text{LT}} \sim 10\text{--}20$ suffices for $10^{-6}\text{--}10^{-7}$ relative accuracy on the denominator. The exchange diagram, at the cost of a conventional RI-MP2 exchange, is the practical ceiling on the full calculation: its $v \times v$ per-pair work is not yet truncated to pair domains, so it is affordable to $v \sim 3000\text{--}4000$. Beyond that, $c_{\text{ss}} = 0$ removes it and the direct term alone scales to the largest spinor bases with a memory footprint of a few K^2 complex numbers held or streamed.

V. VALIDATION

The working equations were checked at three levels.

Kernel identities. On random collocation factors and a random Hermitian core, the compiled direct kernel reproduces the reference evaluation of Eq. (8) to machine precision ($\sim 10^{-15}$ relative), and both reproduce the defining sum $-\frac{1}{2} \sum_{ijab} |(ia|jb)|^2 \sum_g w_g e^{-D_{ijab} t_g}$ built densely from the same factors; the residual against the exact $1/D$ evaluation falls with the Laplace spacing as expected. The out-of-core direct path, with the core handed back in row panels, agrees with the in-core path to $\sim 10^{-14}$. The compiled exchange kernel, summed over the full pair list, reproduces the exact-denominator reference of Eq. (11) to $\sim 10^{-15}$ and the Laplace form of the same quantity to the quadrature accuracy.

Against a dense density-fitted MP2. For a closed-shell atom (neon, X2CAMF, correlation-consistent double- ζ with a matching auxiliary set) the three-index integrals were transformed into the X2CAMF spinor molecular-orbital basis and the antisymmetrised MP2 energy formed densely from Eq. (2). The interpolation-separable factors were then built by the procedure of Sec. III and the kernel run: the exchange diagram matched the dense value to $\sim 10^{-15}$; the direct diagram and the total matched to the tensor-hypercontraction fit and Laplace quadrature error, $\sim 10^{-6}$ – 10^{-7} in the correlation energy at the default grid and spacing. A frozen-core run (2 frozen spinors) reproduced a dense frozen-core RI-MP2 in the same active space to the same accuracy.

Independent paths. The in-core and out-of-core direct evaluations, the compiled and reference kernels, and the scaled-opposite-spin ($c_{ss} = 0$) path were checked pairwise for consistency on the same inputs.

VI. CONCLUSIONS

A memory-bounded MP2 for two-component relativistic references follows from treating the two Goldstone diagrams by different factorisations: the direct diagram by tensor hypercontraction and a Laplace transform, which reduce it to $K \times K$ grid-space matrix multiplications at $\mathcal{O}(K^3)$ per node with $\mathcal{O}(K^2)$ storage; the exchange diagram, which the Laplace transform does not decouple, by the resolution of the identity with the occupied labels explicit, so that it needs no quadrature and is exact to the RI fit. The interpolation factors are built by a grid-blocked ISDF procedure that avoids every grid-squared array, the direct kernel has an out-of-core path that streams the THC core, and the exchange kernel takes a Schwarz pair screen. With the exchange diagram removed the method is a scaled-opposite-spin relativistic MP2 that scales to the largest spinor bases; the remaining obstacle to a full relativistic MP2 at that size is a pair-domain truncation of the exchange, which is left for future work.

Appendix A: Grid THC / ISDF conventions

The spinor pair density on the parent grid is $\rho_{pq}(P) = \sum_{k=1}^2 X_{kPp}^* X_{kPq}$ with $X_{kPp} = w_P^{1/4} \varphi_p^{(k)}(\mathbf{r}_P)$ and $\varphi_p^{(k)}$ the k -th component of spinor p . The least-squares THC core minimises

$\sum_{pqrs} |(pq|rs) - \sum_{PQ} \rho_{pq}(P) Z_{PQ} \rho_{rs}(Q)|^2$ with $(pq|rs)$ taken from the RI fit; its stationary condition is $SZS = E$ with $S_{PQ} = \sum_{pq} \rho_{pq}(P)^* \rho_{pq}(Q)$ and $E_{PQ} = \sum_{pqrs} \rho_{pq}(P) (pq|rs) \rho_{rs}(Q)$, matching the convention already used by the X2CAMF THC coupled-cluster code so that the factors are interchangeable between methods.

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